

AWS

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Foundation

- What is cloud computing
- Regions, Availability Zone
- Networking basics (IP, DNS, HTTP/HTTPS)
- Linux Basics
- Basic Git

AWS → It is just Infrastructure.

AWS Core Services :-

* compute

- ↳ EC2
- ↳ Auto Scaling

* storage

- ↳ S3 (Object Storage)

* Database

- ↳ RDS (SQL DB)
- ↳ Dynamo DB (NoSQL)

* Networking

- ↳ VPC
- ↳ Subnets, Route Tables, Internet Gateway

* Security

↳ IAM (Users, roles, policies).

These are the backbone of every production system.

Build :-

- Host a website using EC2
- store files in S3
- Connect backend to RDS.

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Phase-2

↳ Real Architecture thinking.

Master

- Load Balancer
- Auto Scaling Group
- CloudWatch (monitoring)
- CloudFront (CDN)
- Route 53 (DNS)

Build :- Scalable Web apps

- ALB + Auto Scaling
- S3 + CloudFront
- Route 53 domain

Phase-III

→ Serverless & Modern AWS.

GUI X — code

Master

- Lambda
- API Gateway
- EventBridge
- SQS (Queue)
- SNS (pub/sub)

Phase-4

→ DevOps & Automation

Master

- CI/CD
- Iac
- Docker
- Ecs/EKS

Phase-5

→ Security & Advanced Networking

→ IAM advanced policies

→ VPC deep dive

○ — NAT Gateway

○ — Private Vs Public subnets

→ Security

○ — WAF

○ — Shield

→ Secret Manager

Phase-6

Data, AI & Modern Workload

* Data

- ↳ Redshift
- ↳ Athena

* Streaming

- ↳ Kinesis

* AI

- ↳ Bedrock (LLM on AWS)
- ↳ SageMaker (ML)

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Phase-7

System Design Mastery

* Designing System like

- ↳ Netflix
- ↳ Uber

↳ whatsapp

* Cost optimization

* Trade-offs